If we choose the level 1 at the surface of the liquid (i.e., air-water interface) and the level 2 at a depth 'h' below the surface (as shown in Figure 7.10(b)), then the value of h_1 becomes zero ($h_1 = 0$) and in turn P_1 assumes the value of atmospheric pressure (say P_a). In addition, the pressure (P_2) at a depth becomes P. Substituting these values in equation (7.16), we get

$$P = P_a + \rho gh \qquad (7.17)$$

Therefore, the pressure at a depth h is greater than the pressure on the surface of the liquid.

If the atmospheric pressure is neglected or ignored, then the pressure at a depth *h* is

$$P = \rho gh \qquad (7.18)$$

For a given liquid, ρ is fixed and g is also constant, then the pressure due to the fluid column is directly proportional to vertical distance or height of the fluid column. This implies that the height of the fluid column is more important to decide the pressure and not the cross sectional or base area or even the shape of the container.

Hydrostatic Paradox

When we talk about liquid at rest, the liquid pressure is the same at all points at the same horizontal level (or same depth). This statement can be demonstrated by an experiment called 'hydrostatic paradox'. Let us consider three vessels of different shapes A, B, and C as shown in Figure 7.11. These vessels are connected at the bottom by a horizontal pipe. When they are filled with a liquid (say water), it occupies the same level even though the vessels hold different amounts of water. It is true because the liquid at the bottom of each section of the vessel experiences the same pressure.

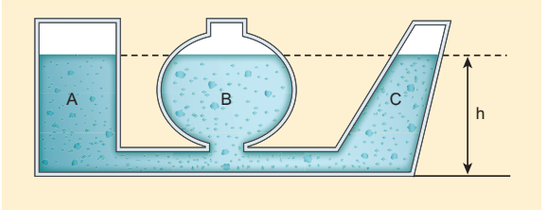


Figure 7.11 Illustration of hydrostatic parodox



The atmospheric pressure at a place is the gravitational force exerted by air above that place per unit surface area. It changes with height and weather conditions (i.e. density of air). In fact, the atmospheric pressure decreases with increasing elevation.

The decrease of atmospheric pressure with altitude has an unwelcome consequence in daily life. For example, it takes longer time to cook at higher altitudes. Nose bleeding is another common experience at higher altitude because of larger difference in atmospheric pressure and blood pressure. Its value on the surface of the Earth at sea level is 1atm.

1 atm pressure = 1.013×10^5 pa

Activity

Take a metal container with an opening. Connect a vacuum pump to the opening. Evacuate the air from inside the container. Why the shape of the metal container gets crumbled?

Inference:

Due to the force of atmospheric pressure acting on its outer surface, the shape of the container crumbles.

UNIT-7(XI-Physics_Vol-2)_14-08-2020.indd 62 17-08-2020 11:31:00